RELATIVE PERFECTNESS DESCENDS ALONG THE APPROXIMATION SYSTEM

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file perfect.tex; no statement or proof step was rewritten, reordered or removed. The counted span is the author environment and its own proof environment. References inside the span are handed to the assembler as navigation stubs that carry the canonical tag of the cited tag only; whether each of them is load-bearing is recorded on the stub itself.

Allowed standard prerequisites (editorial).

- The functor from the colimit of the descent categories to the limit category is fully faithful, as shown for homomorphisms. used at proof lines 9127-9134. Reduces the equivalence to essential surjectivity.
- Objects and maps can be glued along unions of opens from data on the pieces together with an identification on the overlap. invoked at proof lines 9160-9168. Assembles the descent datum for E over U and V into an object E_i over the union.
- The induction principle for quasi-compact quasi-separated schemes allows statements about opens to be reduced to the affine case by successive extensions along unions. invoked at proof lines 9171-9180. Reduces the essential surjectivity to affine X_0 and S_0.
- For an affine limit the colimit of relatively perfect module categories identifies with the relatively perfect modules over the colimit ring. invoked at proof lines 9181-9185. Handles the affine case; the translation is done by the affine locally relatively perfect lemma.

Source reference. Original remark 0DI6: sources/stacks-perfect/pinned-original-source.tex, source lines 9045–9045. Referenced at statement line 9119 for the setup; the gluing and reduction arguments run inside this situation and inside the counted span.

Source reference. Original lemma 0DI7: sources/stacks-perfect/pinned-original-source.tex, source lines 9057–9057. Cited at proof lines 9130-9134 for full faithfulness and again at lines 9150-9156 to enlarge i and at 9171-9174 for the reduction pattern; the essential surjectivity argument is inside the counted span.

Source reference. Original lemma 08DG: sources/stacks-chow/tags.txt, source lines 1–1. Cited at proof lines 9160-9168 to assemble E_i from the pieces; the verification that E_i is S_i-perfect and d is adjoint to an isomorphism is inside the counted span.

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Source reference. Original lemma 08DR: sources/stacks-chow/tags.txt, source lines 1–1. Cited at proof lines 9174-9180 for the reduction to the affine case; the two-stage reduction is inside the counted span.

Source reference. Original lemma 0DHX: sources/stacks-chow/tags.txt, source lines 1–1. Cited at proof lines 9181-9183 for the affine case; the reduction to it is inside the counted span.

Source reference. Original lemma 0DI2: sources/stacks-perfect/pinned-original-source.tex, source lines 8951–8951. Cited at proof lines 9184-9185 for the translation into algebra; the assembly of the affine case is inside the counted span.

Lemma 0.2. *In Situation 0DI6 the category of S-perfect objects of $D(\mathcal{O}_X)$ is the colimit of the categories of S_i-perfect objects of $D(\mathcal{O}_{X_i})$.*

Proof. For every quasi-compact open $U_0 \subset X_0$ consider the condition P that the functor

$$\operatorname{colim}_{i \geq 0} D_{S_i\text{-perfect}}(\mathcal{O}_{U_i}) \longrightarrow D_{S\text{-perfect}}(\mathcal{O}_U)$$

is an equivalence where $U = f_0^{-1}(U_0)$ and $U_i = f_{i0}^{-1}(U_0)$. We observe that we already know this functor is fully faithful by Lemma 0DI7. Thus it suffices to prove essential surjectivity.

Suppose that P holds for quasi-compact opens U_0, V_0 of X_0. We claim that P holds for $U_0 \cup V_0$. We will use the notation $U_i = f_{i0}^{-1}U_0$, $U = f_0^{-1}U_0$, $V_i = f_{i0}^{-1}V_0$, and $V = f_0^{-1}V_0$ and we will abusively use the symbol f_i for all the morphisms $U \to U_i$, $V \to V_i$, $U \cap V \to U_i \cap V_i$, and $U \cup V \to U_i \cup V_i$. Suppose E is an S-perfect object of $D(\mathcal{O}_{U \cup V})$. Goal: show E is in the essential image of the functor. By assumption, we can find $i \geq 0$, an S_i-perfect object $E_{U,i}$ on U_i, an S_i-perfect object $E_{V,i}$ on V_i, and isomorphisms $Lf_i^*E_{U,i} \to E|_U$ and $Lf_i^*E_{V,i} \to E|_V$. Let

$$a : E_{U,i} \to (Rf_{i,*}E)|_{U_i} \quad \text{and} \quad b : E_{V,i} \to (Rf_{i,*}E)|_{V_i}$$

the maps adjoint to the isomorphisms $Lf_i^*E_{U,i} \to E|_U$ and $Lf_i^*E_{V,i} \to E|_V$. By fully faithfulness, after increasing i, we can find an isomorphism $c : E_{U,i}|_{U_i \cap V_i} \to E_{V,i}|_{U_i \cap V_i}$ which pulls back to the identifications

$$Lf_i^*E_{U,i}|_{U \cap V} \to E|_{U \cap V} \to Lf_i^*E_{V,i}|_{U \cap V}.$$

Apply Cohomology, Lemma 08DG to get an object E_i on $U_i \cup V_i$ and a map $d : E_i \to Rf_{i,*}E$ which restricts to the maps a and b over U_i and V_i. Then it is clear that E_i is S_i-perfect (because being relatively perfect is a local property) and that d is adjoint to an isomorphism $Lf_i^*E_i \to E$.

By exactly the same argument as used in the proof of Lemma 0DI7 using the induction principle (Cohomology of Schemes, Lemma 08DR) we reduce to the case where both X_0 and S_0 are affine. (First work with opens in S_0 to reduce to S_0 affine, then work with opens in X_0 to reduce to X_0 affine.) In the affine case the result follows from More on Algebra, Lemma 0DHX. The translation into algebra is done by Lemma 0DI2. □